

HEALTHCONNECT

Appointment Attendance & No-Show Analysis

Week 5 Data Analytics Report

Prepared using Power BI

September 2026

Table of Contents

1. Introduction and Week 4 Review
2. Problem Statement
3. Week 5 Objectives
4. Data Preparation and Cleaning
5. Exploratory Data Analysis
6. KPI Development
7. Dashboard and Visualisations
8. Key Findings and Business Insights
9. Recommendations
10. Assumptions and Limitations
11. Challenges and Decisions
12. Conclusion
13. Proposed Week 6 Focus

1. Introduction and Week 4 Review

HealthConnect is a healthcare appointment management environment that requires analysis of appointment attendance patterns and factors associated with missed appointments. The purpose of this project is to analyse appointment data to identify meaningful patterns in patient attendance, no-shows and cancellations.

During Week 4, the project foundation was established by defining the business problem, identifying the available HealthConnect data and considering an analytical approach. The focus for Week 5 was to move from planning into practical data analysis by preparing the dataset, conducting exploratory analysis, developing key performance indicators (KPIs) and creating an initial analytical dashboard.

The analysis was completed using Power BI, with Power Query used for data inspection and preparation.

During Week 4, the foundation for the HealthConnect project was established by defining the business problem, identifying the relevant data and developing an initial analytical approach. The project focused on understanding appointment attendance and identifying factors that may be associated with no-shows and cancellations.

The planned focus for Week 5 was to move from the project planning stage to practical data analysis. This included inspecting and preparing the HealthConnect dataset, conducting exploratory data analysis, developing key performance indicators and creating an initial analytical dashboard.

During Week 5, these planned activities were completed using Power BI and Power Query. The analysis resulted in the development of five KPIs, multiple analytical visualisations and initial business insights relating to appointment outcomes, demographics, appointment characteristics, reminder channels, waiting time, distance to the clinic and previous no-show history.

2. Problem Statement

HealthConnect needs to better understand appointment attendance and the high occurrence of missed appointments. Missed appointments can negatively affect healthcare operations because appointment slots may remain unused while other patients could potentially have used those slots.

The primary analytical question for this project was:

What patterns exist in HealthConnect appointment attendance and no-shows, and which patient, appointment and operational factors may be associated with appointment outcomes?

3. Week 5 Objectives

- Prepare and inspect the HealthConnect appointment dataset.
- Review data quality and identify potential data issues.
- Conduct exploratory data analysis.
- Develop meaningful KPIs relating to appointment attendance.
- Analyse appointment outcomes across different variables.
- Create an interactive Power BI dashboard.
- Identify meaningful business insights.
- Provide initial recommendations for HealthConnect.

4. Data Preparation and Cleaning

The HealthConnect appointment dataset was imported into Power BI and examined using Power Query. The data preparation process included reviewing column names, data types, missing values, errors, data completeness, potential duplicate records and consistency of categorical values.

The dataset contained 5,000 appointment records.

Data Quality Checks

Column profiling and distribution tools in Power Query were used to inspect the dataset. Variables reviewed included:

- Appointment ID and Patient ID
- Gender, Age and Age Group
- Appointment Type
- Booking Date and Appointment Date
- Appointment Day and Appointment Time
- Booking Lead Days
- Previous Appointments and Previous No-Shows
- Reminder Sent and Reminder Channel
- Distance to Clinic and Waiting Time
- Appointment Outcome

Data quality indicators were reviewed to assess valid values, errors and empty values. Data types were also checked to ensure that variables were suitable for analysis.

5. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the appointment data and identify patterns in appointment outcomes.

- Overall appointment outcomes
- No-shows, attended appointments and cancellations
- Age groups and gender
- Appointment day and appointment type
- Reminder channels
- Previous no-shows
- Waiting time
- Distance to clinic

6. KPI Development

Five KPIs were developed for the HealthConnect dashboard.

Total Appointments

Definition: The total number of appointments recorded in the dataset.

Result: 5,000

Why it matters: It provides the overall volume of appointments and forms the baseline for analysing attendance, cancellations and no-shows.

Business question: How many appointments were recorded during the period covered by the dataset?

Interpretation: The dataset contains **5,000 appointments**, providing the population used for this analysis.

No-Shows

Definition: The number of appointments where the patient did not attend.

Result: 2,423

Why it matters:

No-shows can result in unused appointment capacity and may negatively affect operational efficiency.

Business question:

How many appointments resulted in patients not attending?

Interpretation:

A total of **2,423 appointments** resulted in no-shows, indicating that missed appointments are a significant issue in the dataset.

Attended Appointments

Definition: The number of appointments successfully attended by patients.

Result: 2,314

Why it matters:

It measures successful appointment attendance and helps HealthConnect understand how many scheduled appointments resulted in patient engagement.

Business question:

How many appointments were successfully attended?

Interpretation:

A total of **2,314 appointments** were attended.

Cancelled Appointments

Definition: The number of appointments that were cancelled.

Result: 263

Why it matters:

Cancellations are important because they represent another appointment outcome and may differ operationally from no-shows.

Business question:

How many appointments were cancelled?

Interpretation:

A total of **263 appointments** were cancelled, making cancellations considerably less common than both attended appointments and no-shows.

No-Show Rate

Definition: The percentage of total appointments that resulted in a no-show.

Result: 48.46%

No-Show Rate Calculation

No-Show Rate = (Number of No-Shows ÷ Total Appointments) × 100

Why it matters:

The no-show rate allows HealthConnect to assess the scale of missed appointments relative to the total number of appointments.

Business question:

What proportion of scheduled appointments resulted in a no-show?

Interpretation:

The no-show rate is **48.46%**, meaning that almost half of all appointments in the dataset resulted in a patient not attending.

7. Dashboard and Visualisations

An interactive Power BI dashboard titled “HealthConnect Appointment Attendance & No-Show Analysis” was developed to provide a high-level overview and detailed analysis of appointment outcomes.

- Appointment Outcome Distribution
- No-Shows by Age Group
- Outcomes by Appointment Day
- Outcomes by Gender
- Outcomes by Appointment Type
- Outcomes by Reminder Channel
- Average Waiting Time by Appointment Outcome
- Previous No-Shows
- Average Distance to Clinic by Appointment Outcome

8. Key Findings and Business Insights

Insight 1: HealthConnect has a high no-show rate

Out of 5,000 appointments, 2,423 resulted in no-shows. This represents a no-show rate of 48.46%. Almost one out of every two appointments in the dataset resulted in the patient not attending.

Business implication: The high proportion of no-shows may affect operational efficiency and the utilisation of appointment capacity. HealthConnect should prioritise further investigation into factors associated with missed appointments.

Insight 2: No-shows slightly exceed attended appointments

The analysis showed 2,423 no-shows, 2,314 attended appointments and 263 cancelled appointments. No-shows slightly exceeded the number of attended appointments.

Business implication: Appointment attendance is a significant operational concern within the dataset.

Insight 3: Appointment outcomes vary across demographic groups

The dashboard shows differences in appointment outcomes across age groups and gender.

Business implication: Demographic information may assist HealthConnect in identifying groups requiring further analysis. Outcome rates should be calculated before concluding that a specific group has a higher no-show risk.

Insight 4: Appointment outcomes differ across appointment characteristics

Appointment outcomes vary across appointment days and appointment types, including General Consultation, Follow-up, Specialist Consultation and Diagnostic Test.

Business implication: Different appointment types may have different attendance patterns and may require different scheduling or communication strategies.

Insight 5: Reminder channels show broadly similar outcome patterns

Appointment outcomes were compared across SMS, WhatsApp, Email and no reminder. The visual analysis showed broadly similar outcome patterns across reminder channels.

Business implication: This does not prove reminders are ineffective. Further analysis should investigate reminder timing, frequency, content, delivery success and patient communication preferences.

Additional Analytical Findings

Waiting Time

Average waiting time appeared relatively similar across attended appointments, no-shows and cancelled appointments. The differences were small, suggesting waiting time does not appear to be a strong differentiating factor in this initial analysis.

Distance to Clinic

Average distance to the clinic was relatively similar across no-shows, cancelled appointments and attended appointments. Distance does not appear to be a major differentiating factor in this initial analysis.

Previous No-Shows

The largest number of appointments involved patients with few or no previous no-shows, while appointment counts decreased as previous no-shows increased. The current visual shows counts rather than no-show rates, so it cannot conclude that previous no-shows predict future missed appointments.

9. Recommendations

Investigate the high no-show rate

Conduct further analysis into patient scheduling behaviour, appointment accessibility, reminder timing, appointment confirmation and patient history.

Analyse no-show rates by category

Calculate no-show rates, rather than relying only on appointment counts, across age groups, gender, appointment type, appointment day and reminder channel.

Evaluate reminder effectiveness

Investigate whether reminder timing, frequency or content can be improved and whether patients have different communication preferences.

Use appointment history for further risk analysis

Analyse no-show rates for patients with zero, one or multiple previous no-shows.

Continue monitoring operational factors

Continue analysing waiting time and distance to clinic because averages may hide variation between individual patients.

10. Assumptions and Limitations

Week 5 Assumptions, Limitations, Risks and Dependencies

The assumptions and limitations identified during Week 4 remained relevant during Week 5. The analysis continued to depend on the completeness, accuracy and availability of the HealthConnect appointment dataset.

Assumptions

The analysis assumes that:

- The appointment records accurately represent the HealthConnect appointment activity included in the dataset.
- Appointment outcomes are correctly recorded as Attended, No-Show or Cancelled.
- Patient and appointment information is sufficiently accurate for analysis.
- Variables such as waiting time, reminder channel and distance to clinic are recorded consistently.

Limitations

Some limitations remained relevant during Week 5. The analysis was limited to the variables available in the dataset and did not include external factors that may influence appointment attendance.

Examples include:

- Transport availability

- Patient financial circumstances
- Weather conditions
- Medical urgency
- Personal circumstances

Risks

A key analytical risk is interpreting appointment counts as evidence of higher no-show risk. Categories with more appointments may naturally have higher numbers of no-shows.

To address this, future analysis should calculate and compare **no-show rates** across categories rather than relying only on counts.

Dependencies

The analysis depended on the availability of complete and reliable appointment data. The accuracy of the dashboard findings and business insights is dependent on the quality of the underlying dataset.

Future analysis may also depend on additional information or collaboration with other analytical areas to gain a deeper understanding of factors influencing patient attendance.

Dataset limitation

The analysis is based only on the available dataset and does not include external factors such as transport availability, weather, appointment urgency or other contextual information.

Counts do not always indicate risk

A larger number of no-shows does not automatically mean that a category has a higher no-show risk because categories may contain different numbers of appointments.

Association does not prove causation

The analysis identifies patterns and relationships but cannot prove that one factor causes another.

Averages may hide variation

Average waiting time and average distance may not capture differences between individual patients. Further analysis could examine distributions, ranges and outliers.

11. Challenges and Decisions

Power BI Performance

During the data preparation process, Power BI experienced freezing and performance issues. The application was closed and restarted before continuing with the analysis.

Data Preparation

Column profiling was used to inspect data quality, including valid values, errors and empty values.

Dashboard Design

The dashboard was designed to provide a high-level overview through KPI cards and detailed analysis through interactive visualisations. Consistent formatting and chart layouts were used to improve readability.

12. Conclusion

The Week 5 analysis provided an initial understanding of HealthConnect appointment attendance and no-show patterns. The analysis found that 48.46% of appointments resulted in no-shows, representing a significant proportion of the 5,000 appointments analysed.

The Power BI dashboard provided insights into appointment outcomes across demographic, appointment and operational variables. The findings suggest that HealthConnect should prioritise further investigation into factors associated with no-shows.

The work completed during Week 5 provides a foundation for further analysis, improved risk assessment and continued development of the HealthConnect analytics project.

13. Proposed Week 6 Focus

- Further investigation of no-show risk factors.
- Calculation of no-show rates across different categories.
- Deeper analysis of patient appointment history.
- Improvement of dashboard interactivity.
- Identification of more targeted business recommendations.
- Refinement and documentation of analytical findings.

14. Week 5 Project Summary

During Week 5, the HealthConnect project moved from the planning and problem-definition stage into practical data analysis.

The planned activities for the week included inspecting and preparing the dataset, conducting exploratory data analysis, developing key performance indicators and creating an initial Power BI dashboard. These activities were completed using Power BI and Power Query.

The dataset containing **5,000 appointment records** was reviewed for data quality and suitability for analysis. The analysis focused on appointment outcomes, patient demographics, appointment characteristics, reminder channels, previous no-show history, waiting time and distance to the clinic.

Five key performance indicators were developed:

- Total Appointments
- No-Shows
- Attended Appointments
- Cancelled Appointments

- No-Show Rate

One of the most significant findings was the **48.46% no-show rate**, with **2,423 appointments resulting in no-shows**. No-shows also slightly exceeded attended appointments, highlighting appointment attendance as an important area for further investigation.

An interactive Power BI dashboard was developed to visualise the findings and support the identification of patterns across different variables.

A challenge encountered during the analysis was Power BI freezing during the data preparation process. This was addressed by restarting the application and continuing the analysis.

An important analytical decision was to interpret findings carefully and avoid treating appointment counts as evidence of higher no-show risk. Future analysis should therefore focus on calculating no-show rates across categories.

The main remaining work includes deeper analysis of no-show risk factors, further examination of appointment history and continued refinement of the dashboard and business recommendations.

The proposed focus for Week 6 is to build on the Week 5 findings by conducting deeper analysis and identifying more targeted insights and recommendations.

15. Cross-Track Collaboration and Dependencies

The Week 5 Data Analytics work creates a foundation for other areas of the HealthConnect project. The appointment data, KPIs and analytical findings can provide useful inputs for further work in related tracks.

Data Analytics and Data Science

The exploratory analysis and KPIs developed during Week 5 can support future Data Science work. Variables identified during the analysis, such as appointment type, previous no-shows, reminder channel, waiting time and distance to the clinic, may be useful when investigating factors associated with appointment outcomes.

The Data Analytics findings can therefore help identify potential variables for further predictive or advanced analysis.

Project Dependency

The Data Analytics track depends on reliable and complete underlying data. Any future changes to the dataset or the availability of additional variables may require the analysis and dashboard to be updated.

Similarly, future analytical or Data Science findings may provide additional insights that can improve the HealthConnect dashboard and business recommendations.

16. Power BI Dashboard

The Power BI dashboard was developed to provide a visual summary of the HealthConnect appointment data and support the identification of patterns relating to appointment attendance, no-shows and cancellations. The dashboard includes key performance indicators and visualisations covering appointment outcomes, demographics, appointment characteristics, reminder channels, waiting time, previous no-show history and distance to the clinic.

Figure 1: HealthConnect Appointment Attendance and No-Show Analysis Dashboard

